

Series of real numbers

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots, \quad a_n \in \mathbb{R}, \quad \forall n \geq 1$$

$$(s_n), \quad n \geq 1 \Rightarrow s_n = a_1 + a_2 + \dots + a_n$$

$$\exists \lim s_n = s \in \overline{\mathbb{R}}$$

s = sum of the series

$s \in \mathbb{R} \Rightarrow$ convergent series

Th. 1. (Cauchy's general test of convergence)

$$\sum_{n=1}^{\infty} a_n \text{ is convergent} \Leftrightarrow \forall \varepsilon > 0, \exists n_{\varepsilon} \in \mathbb{N} \text{ s.t.}$$

$$\forall n \geq n_{\varepsilon}, \forall p \in \mathbb{N}:$$

$$|a_{n+1} + a_{n+2} + \dots + a_{n+p}| < \varepsilon$$

Proof: $\sum_{n=1}^{\infty} a_n$ conv. $\Leftrightarrow s_n$ conv. $\Leftrightarrow (s_n), n \geq 1$

Fundamental $\Leftrightarrow \varepsilon > 0, \exists n_{\varepsilon} \in \mathbb{N}$ s.t. $\forall n \geq n_{\varepsilon}, \forall p \in \mathbb{N}:$

$$|s_{n+p} - s_n| < \varepsilon \Leftrightarrow |a_{n+1} + \dots + a_{n+p}| < \varepsilon, \quad \forall n \geq n_{\varepsilon}$$

Th. 2. (A necessary condition of convergence) IF

$$\sum_{n=1}^{\infty} a_n \text{ is convergent} \Rightarrow \lim_{n \rightarrow \infty} a_n = 0$$

Proof: $a_n = (a_1 + a_2 + \dots + a_n) - (a_1 + \dots + a_{n-1}) =$

$$s_n - s_{n-1}$$

$$\sum_{n=1}^{\infty} a_n \text{ conv.} \Rightarrow \exists \lim_{n \rightarrow \infty} r_n = r \in \mathbb{R}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (r_n - r_{n-1}) = r - r = 0$$

Obs. 1) The condition $\lim a_n = 0$ is not sufficient for convergence. (Not $\Leftarrow$)

$$\text{ex. } \sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots = +\infty$$

$$\text{and } \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$2) \text{ If } \lim a_n \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n \text{ is divergent}$$

$$\text{ex. } \sum_{n=1}^{\infty} \frac{n}{n+1} = +\infty \quad \left(\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \neq 0 \right)$$

1. Series with positive terms

$$\sum_{n=1}^{\infty} a_n, \quad a_n > 0, \quad \forall n \geq 1$$

$(r_n), n \geq 1$ is strictly increasing

Def. $\sum_{n=1}^{\infty} a_n$ is conv. $\Leftarrow (r_n), n \geq 1$ is bounded.

Th. 1 (First comparison test). Let $\sum_{n=1}^{\infty} a_n, \sum_{n=1}^{\infty} b_n$

be series with positive terms s.t. $a_n \leq b_n, \forall n \geq n_0$.

Then: 1) If $\sum_{n=1}^{\infty} b_n$ is conv. $\Rightarrow \sum_{n=1}^{\infty} a_n$ is conv.

2) If $\sum_{n=1}^{\infty} a_n$ in div. $\Rightarrow \sum_{n=1}^{\infty} b_n$ in div.

Proof: We may suppose without loss of generality that $n_0 = 1$, because if we eliminate a finite number of terms, it's nature remains the same.

$$\begin{aligned} n_n &= a_1 + a_2 + \dots + a_n \\ t_n &= b_1 + b_2 + \dots + b_n \end{aligned} \quad \left\{ \begin{aligned} a_n &\leq b_n \\ &= \end{aligned} \right. \quad n_n \leq t_n, \forall n \geq 1$$

1) $\sum_{n=1}^{\infty} b_n$ conv. $\Rightarrow t_n$ is convergent $\Rightarrow (t_n)$ is bounded $\Rightarrow (n_n)$ is bounded $\Rightarrow \sum_{n=1}^{\infty} a_n$ is conv.

2) $\sum_{n=1}^{\infty} a_n = +\infty \Rightarrow n_n \rightarrow \infty \Rightarrow t_n \rightarrow \infty \Rightarrow \sum_{n=1}^{\infty} b_n$ is divergent.

ex. Test the conv. of the series:

$$\sum_{n=1}^{\infty} \frac{n}{n^3+1}, \quad a_n = \frac{n}{n^3+1}, \quad n \geq 1$$

$$a_n \leq \frac{n}{n^3} = \frac{1}{n^2} = b_n, \quad \forall n \geq 1$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \text{ (convergent)} \quad \left| \Rightarrow \sum_{n=1}^{\infty} a_n \text{ conv.} \right.$$

Th. 2. (Second comparison test) Let $\sum_{n=1}^{\infty} a_n, \sum_{n=1}^{\infty} b_n$ be series with positive terms n.f.:

$$\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}, \quad \forall n \geq n_0$$

Then: 1) If $\sum_{n=1}^{\infty} b_n$ is conv $\Rightarrow \sum_{n=1}^{\infty} a_n$ conv.

2) If $\sum_{n=1}^{\infty} a_n$ is div. $\Rightarrow \sum_{n=1}^{\infty} b_n$ is div.

Th. 3. (Third comparison test) Let $\sum_{n=1}^{\infty} a_n, \sum_{n=1}^{\infty} b_n$ be series with positive terms n.t.:

$$\exists \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \ell, \ell \in [0, +\infty)$$

Then: 1) If $\ell \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n, \sum_{n=1}^{\infty} b_n$ have the same nature (both conv. / div.)

2) If $\ell = 0$ and $\sum_{n=1}^{\infty} b_n$ is div. $\Rightarrow \sum_{n=1}^{\infty} a_n$ is div.

ex. Test the conv. of the series:

$$\sum_{n=1}^{\infty} \frac{\arctg n}{n} = \frac{\arctg 1}{1} + \frac{\arctg 2}{2}$$

$$a_n = \frac{\arctg n}{n} \text{ Compare to } \sum_{n=1}^{\infty} \frac{1}{n} = +\infty, b_n = \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \arctg n = \frac{\pi}{2} \neq 0 \xrightarrow{\text{Th. 3}} \sum_{n=1}^{\infty} a_n = +\infty$$

Obs. A consequence of the comparison test is:

Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms n.t.

$\alpha \in \mathbb{R}$:

$$\lim_{n \rightarrow \infty} n^{\alpha} \cdot a_n = \ell, \ell \in [0, +\infty)$$

Then: 1) If $\alpha > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is conv.

2) If $\alpha \leq 1$ and $\ell \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ is div.

Proof: $\lim_{n \rightarrow \infty} \frac{a_n}{\frac{1}{n^{\alpha}}} = \ell$ and compare to $\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}}$

$\Rightarrow$ The conclusion in the conseq. of the third conseq. test.

ex. Test the conv. of the series:

$$\sum_{n=1}^{\infty} (\sqrt{n^2+1} - n)$$

$$a_n = \sqrt{n^2+1} - n = \frac{1}{\sqrt{n^2+1} + n}$$

$$\lim n \cdot a_n = \lim \frac{n}{\sqrt{n^2+1} + n} = \frac{1}{2} \stackrel{L=1}{\Rightarrow} \sum_{n=1}^{\infty} a_n = +\infty$$

Th. 5: (Cauchy's Root test) Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms n.t.:

$$\exists \lim \sqrt[n]{a_n} = \ell, \ell \in \overline{\mathbb{R}}$$

Then: 1) If $\ell < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent

2) If $\ell > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is divergent

3) If $\ell = 1 \Rightarrow$ The test fails

ex. $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2}$ convergence?

$$a_n = \left(\frac{n}{n+1}\right)^{n^2} = \left(\left(\frac{n}{n+1}\right)^n\right)^n \rightarrow 0$$

$$\lim \sqrt[n]{a_n} = \lim \left(\frac{n}{n+1}\right)^n = \frac{1}{e} < 1 \Rightarrow \sum_{n=1}^{\infty} a_n \text{ conv.}$$

Th. 6 (D'Alembert Ratio Test) Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms n.t.:

$$\lim \frac{a_{n+1}}{a_n} = \rho, \rho \in \overline{\mathbb{R}}$$

Then: 1) If $\rho < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is conv.

2) If $\rho > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is div

3) If $\rho = 1 \Rightarrow$ Test fails

ex. $\sum_{n=1}^{\infty} \frac{\pi^n \cdot n!}{n^n}$ conv.?

$$a_n = \frac{\pi^n \cdot n!}{n^n} \Rightarrow \lim \frac{a_{n+1}}{a_n} = \frac{\pi^{n+1} \cdot (n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{\pi^n \cdot n!}$$

$$= \lim \frac{\pi \cdot (n+1)}{n+1} \cdot \left(\frac{n}{n+1}\right)^n = \frac{\pi}{e} > 1 \Rightarrow \sum_{n=1}^{\infty} a_n \text{ is div.} = +\infty$$

Th. 7: (Raabe - Duhamel Test) Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms n.t.:

$$\exists \lim n \left(\frac{a_n}{a_{n+1}} - 1 \right) = \rho \in \overline{\mathbb{R}}$$

Then: 1) If $\rho < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is divergent

2) If $\rho > 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent

3) If $\rho = 1 \Rightarrow$ Test fails

ex. Study the conv. of the series:

$$\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \dots \cdot 2n}$$

$$a_n = \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \dots \cdot 2n}$$

Ratio test: $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{2n+1}{2n+2} = 1$, test fails

Raabe-Duhamel: $\lim_{n \rightarrow \infty} n \left(\frac{a_n}{a_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{2n+2}{2n+1} - 1 \right)$
 $= \lim_{n \rightarrow \infty} \frac{n}{2n+1} = \frac{1}{2} < 1 \Rightarrow \sum_{n=1}^{\infty} a_n$ is divergent $= +\infty$

Th. 8: (Cauchy's condensation test) Let (a_n) be a decreasing sequence of positive numbers. Then:

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots$$

and

$$\sum_{n=0}^{\infty} 2^n \cdot a_{2^n} = a_1 + 2a_2 + 4a_4 + \dots$$

have the same nature. $\left(\sum_{n=1}^{\infty} a_n \sim \sum_{n=0}^{\infty} 2^n \cdot a_{2^n} \right)$

ex. Prove that $\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$, $\alpha \in \mathbb{R}$, is conv. $\Leftrightarrow \alpha > 1$

$$a_n = \frac{1}{n^\alpha}, \quad \lim_{n \rightarrow \infty} a_n = \begin{cases} \infty, & \alpha < 1 \\ 0, & \alpha = 0 \end{cases} \Rightarrow$$

$\sum_{n=1}^{\infty} a_n$ is divergent

If $\alpha > 1$, then $a_n \searrow 0 \Rightarrow \sum_{n=1}^{\infty} a_n \sim \sum_{n=0}^{\infty} 2^n \cdot a_{2^n}$

$$\sum_{n=0}^{\infty} 2^n \cdot a_{2^n} = \sum_{n=0}^{\infty} 2^n \cdot \frac{1}{(2^n)^\alpha} = \sum_{n=0}^{\infty} (2^n)^{1-\alpha} =$$

$$\sum_{n=0}^{\infty} (2^{1-\alpha})^n = 1 + 2^{1-\alpha} + 2^{(1-\alpha)^2} + \dots \text{ (geometric series)} \Rightarrow$$

$$q = 2^{1-\alpha}, \text{ which is conver. for } q = 2^{1-\alpha} < 1 \Rightarrow \alpha > 1$$

$\Rightarrow$ The GHS is convergent $\Rightarrow \underline{\alpha > 1}$